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Impacts of pretreatment routes on spent lithium-ion batteries recycling

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The massive production and consumption of lithium-ion batteries (LIBs) necessitate the development of end-of-life (EoL) recycling strategies to shape a sustainable LIBs life cycle. Pretreatment, which converts spent LIBs into black mass, represents a foundational stage in the recycling processes. However, our understanding of the impacts of pretreatment on the overall recycling performance and associated environmental footprint is limited. Here we develop operational unit-level models to evaluate the impacts of different pretreatment routes on industrial-scale LIB recycling. The pretreatment routes covered here include mechanical, thermal and chemical routes. We identify the mechanisms driving variability in materials recovery and environmental impacts across different pretreatment routes, cathode chemistries and regional contexts. Our high-resolution analysis reveals that pretreatment accounts for most materials losses and contributes to 16–38% of the environmental impact of LIBs recycling. By identifying key impact contributors, we further propose strategies for targeted process upgrade that could halve environmental impacts during the pretreatment stage. Additionally, this research provides comprehensive process inventories and bills of materials, facilitating further research initiatives, policy design and technology advancement in EoL LIB management, contributing to achieving circular economy in the LIBs sector.

Driven by the accelerating global demand for technologies such as electric vehicles (EVs) and energy storage systems (ESSs) to meet energy transition goals[1], lithium-ion battery (LIB) production has grown rapidly over the past decades[2,3] and is expected to continue rising[4–6]. For example, global LIB demand for EVs increased 11-fold (50–550 GWh yr^{-1}) during 2016–2022, and is projected to reach 3,000–6,000 GWh by 2030[7]. However, alongside this growth, retired battery volumes are also steadily rising, presenting critical challenges for sustainable end-of-life (EoL) management[8]. Landfilling not only forfeits materials recovery (for example, nickel and cobalt) but also poses environmental risks (for example, toxins leaching into ecosystems)[9], prompting many countries to promote LIB recycling as a more responsible solution[10]. This 'urban mining' approach supports a circular economy by reducing reliance on virgin resources, mitigating mineral scarcity and securing supply chains vulnerable to geopolitical tensions[11]. LIB recycling is generally classified into three primary pathways: pyrometallurgical, hydrometallurgical and direct cathode recycling[1,12]. Among these, hydrometallurgical and direct recycling pathways require an essential preliminary stage known as pretreatment, which processes spent LIB cells into black mass, a concentrated mixture of cathode and anode materials[13,14]. As hydrometallurgical processes see growing industrial adoption[15] and direct cathode recycling gradually matures into a more closed-loop alternative[16], pretreatment has become increasingly pivotal for ensuring efficient downstream recovery while minimizing overall operational costs and environmental impacts throughout the entire LIB recycling pathway.

Many review articles have broadly categorized the LIB pretreatment stage into three distinct routes—mechanical, thermal and chemical—based on their fundamental mechanisms[13,14], and described the operation principles and working conditions of key units[17,18].

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The mechanical route relies primarily on physical impact and abrasion (for example, crushing and grinding) to separate metal foils from the black mass[19], and has been widely implemented by industry[20,21] (for example, AkkuSer and Duesenfeld). The thermal route uses high temperatures to decompose the polyvinylidene fluoride (PVDF) binder and enable delamination[22], and has also reached commercial application[21,23] (for example, Accurec and Brunp). By contrast, chemical pretreatment remains largely at the laboratory stage[14]. It uses chemical agents to separate materials while preserving the cathode structure[24,25], including organic solvents (for example, *N*-methyl-2-pyrrolidone, NMP[13]) and inorganic solutions (for example, alkaline media[26]). Recent interest has grown in applying greener solvents[27], such as deep eutectic solvents (DES), because of reduced toxicity and improved biodegradability[25]. Although challenged by reagent cost and operational complexity[1,14], chemical pretreatment remains attractive for its high recovery efficiency and material quality preservation[25].

Apart from descriptive reviews, a growing body of research has applied life-cycle assessment (LCA) to quantify environmental impacts for broader scopes that include LIB pretreatment. However, limited modelling granularity, particularly when assessing full life cycles, typically leads LIB recycling to be represented as a system[28,29] or several stages[30–32], where the pretreatment stage is often inexplicitly modelled in a simplified or generalized form. While some studies have mapped recycling pathways at the process unit level using bill-of-materials (BoM) datasets derived from industry or literature, their analyses remain at the system[33–37] or stage level[30,38,39] because of limited data resolution. A few studies have generated first-hand BoM through unit-level simulations for specific one[40,41] or several[42,43] full recycling pathways, but the environmental footprint behind individual unit operations is not thoroughly revealed, and the modelling or interpretation of pretreatment route variants remains insufficient. Moreover, industrial factors, such as route-specific exhaust treatment, emission species profiles and plant configurations, are rarely incorporated in existing models. Limited consideration of cathode chemistry variations and regional contexts further constrains practical applicability. Overall, given the diversity of pretreatment routes with distinct operational mechanisms, the current modelling and analysis gap hinders understanding of their variability in materials recovery and environmental footprint, thereby limiting insights into the impacts of the pretreatment stage on the entire LIB recycling.

In this study, we develop high-resolution models of the LIB pretreatment stage at the level of individual operational units, designed to reflect industrial-scale practices. The pretreatment framework covers representative mechanical, thermal and chemical routes, each paired with appropriate exhaust treatment systems. Every unit operation is parameterized using data from industry reports, academic literature and practical experience. Mass and energy balance simulations are performed across each unit to allow detailed tracking of material flows and associated environmental burdens. Our bottom-up modelling approach unravels the mechanisms that drive differences in materials recovery and environmental impacts across pretreatment routes, cathode chemistries and regional contexts. It also identifies the key contributors to environmental impacts and quantifies potential improvements achievable through targeted process upgrades. These findings highlight the critical role of the pretreatment stage in shaping the overall sustainability of recycling systems. Furthermore, the robust, accurate data generated in this work could provide a valuable foundation for future research, evidence-based policy-making and technology development in the LIB recycling sector, helping guide the LIB life cycle towards a circular economy.

Results

Pretreatment accounts for most material loss

Figure 1 illustrates the process roadmap for mechanical, thermal and chemical pretreatment of spent LIB cells. Mechanical pretreatment is the simplest in terms of equipment, relying on sequential physical size reduction (for example, crushing and grinding) to delaminate black mass from current collectors (see Supplementary Table 1 for detailed operational unit modelling). Thermal pretreatment involves the fewest process units, wherein shredded LIB cells are calcinated at ~650 °C to decompose the PVDF binder and enable delamination. Chemical pretreatment represents the most procedurally sophisticated methodology of the three routes. Among various applicable reagents, this study selects NMP solvent as a representative option because of its early, traditional and well-known use[14,26], which solubilizes PVDF under ultrasonic and thermal activation conditions. Figure 2a delineates the industrial-scale material flow of black mass, metals, plastics and electrolyte solvents from spent NMC811 cells across each process unit, where NMC811 is selected for analysis as a result of its rapid growth in the LIB market[44] (see Supplementary Tables 4–6 for other cathode chemistries). While all three pretreatment routes enable black mass and metal scrap recovery, mechanical pretreatment produces metal scraps containing the highest black mass impurities as a result of its lower delamination efficiency inherent in the physical separation mechanisms (Supplementary Table 1). Plastics are recoverable in mechanical and chemical routes, whereas thermal pretreatment decomposes them into volatile organic compounds (VOCs) in its high-temperature operations. The chemical route demonstrates superior circularity potentials by achieving a ~92% recovery rate of NMP solvent via condensation and distillation. The key processes concurrently facilitate the recovery of electrolyte solvents from spent LIBs while substantially reducing system discharges.

High-resolution material flow analysis quantifies the material loss rates and product impurities across pretreatment routes for NMC811 cells (Fig. 2b). Critical metals (Ni, Co, Mn and Li) exhibit loss rates ranging from 5.2% to 12.5% during the initial pretreatment phase that transforms LIB cells into black mass. Subsequent downstream metal extraction processes, which convert black mass to metal salts, introduce additional material losses of ~8% for Ni, Co and Mn and ~15% for Li. These findings are critical as they indicate that pretreatment operations are responsible for approximately half the total critical metal losses throughout the entire hydrometallurgical recycling pathway. Among the pretreatment routes evaluated, mechanical pretreatment exhibits the highest critical metal (Ni, Co, Mn and Li) loss of 12.5% and black mass impurity of 8.7%, while thermal and chemical pretreatments show superior performance, with lower loss rates of 7.8% and 5.2%, and impurity rates of 3.9% and 5.2%, respectively. Mechanical pretreatment also suffers from elevated copper losses (6.8%) and impurity levels in metal scraps (26.7%). By contrast, chemical pretreatment achieves better recovery efficiencies, with lower copper loss (5.3%) and impurity content (11.8%). Additionally, the chemical route enables the recovery of electrolyte solvents, with dimethyl carbonate (DMC) and ethylene carbonate (EC) recovered at efficiencies of 87.5% and 70%, respectively. Overall, conventional pretreatment routes (mechanical and thermal) result in material losses of ~30%, whereas chemical pretreatment limits these losses to ~14%, highlighting its superior potentials for advancing material circularity. Given that the subsequent metal extraction stage typically incurs lower losses of ~12%, it becomes evident that the pretreatment stage is a critical determinant of the overall recovery efficiency in LIB recycling pathways.

A notable share of footprint stems from pretreatment

To assess the operational environmental impacts of recycling activities, this study uses a cutoff LCA boundary that excludes both avoided impacts from recovered materials (for example, black mass) and upstream burdens embodied in input recyclables (spent LIB cells). As shown in Fig. 3a, hydrometallurgical recycling is analysed across three endpoint damage categories: human health, ecosystems and resources. Results indicate that human health impacts dominate,

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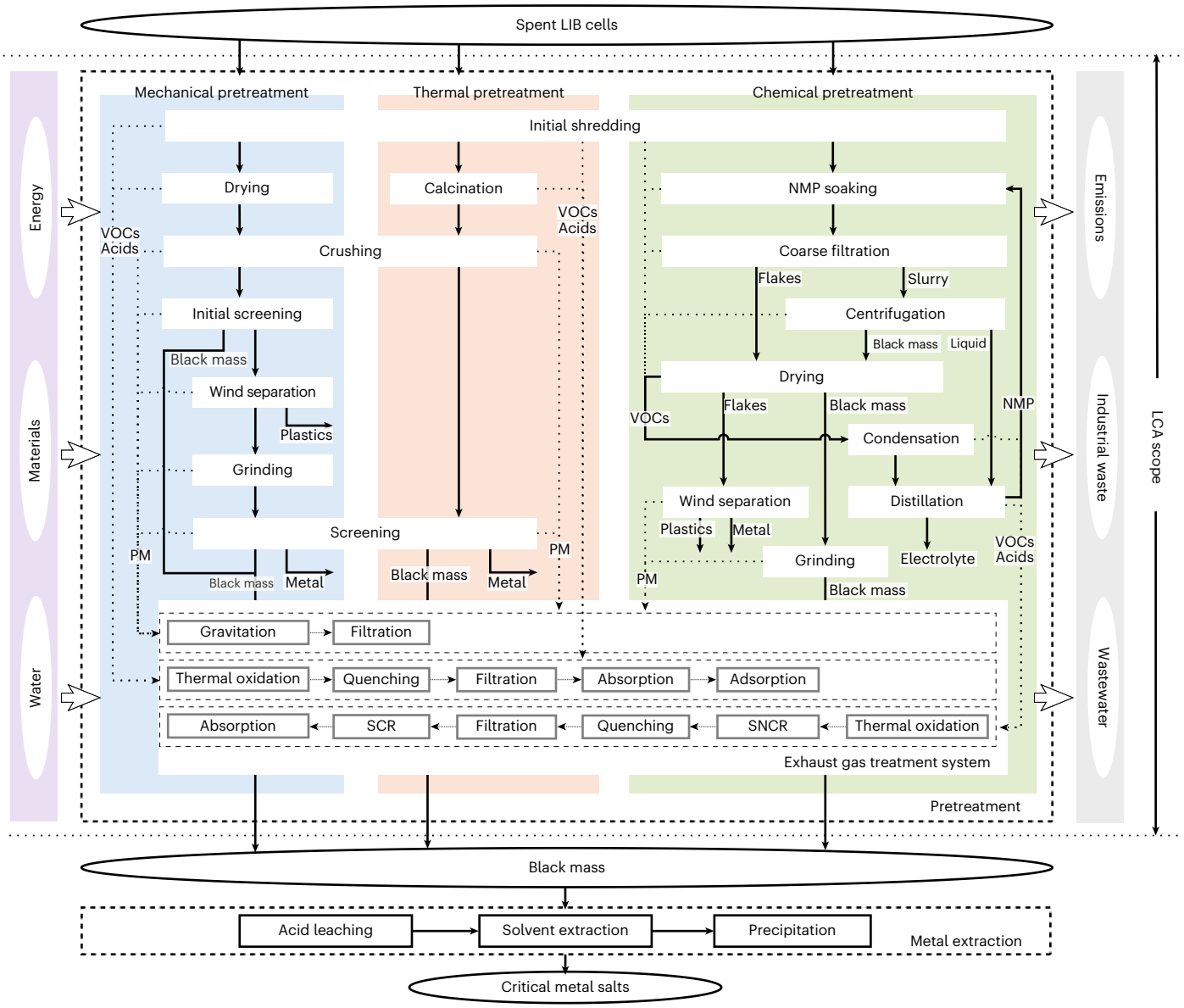


Fig. 1 | Overview of pretreatment configurations for spent LIB cells. The pretreatment stage converts spent LIB cells into black mass, which is subsequently processed into critical metal salts during downstream metal extraction; together, these stages constitute the hydrometallurgical recycling pathway. The pretreatment framework comprises mechanical, thermal and chemical routes, each integrated with route-specific exhaust gas treatment systems. LCA in this study adopts a cutoff boundary to evaluate operational environmental impacts across pretreatment routes, accounting for system inputs (for example, energy, raw materials and water) and output waste streams (for example, emissions, industrial waste and wastewater), while excluding avoided impacts from recovered black mass and byproducts as well as upstream burdens embodied in spent LIB cells. SCR, selective catalytic reduction; SNCR, selective non-catalytic reduction.

accounting for over 96% of total endpoint damage, and are therefore selected as the primary indicator for further footprint analysis. Disaggregating the full recycling pathway into pretreatment and metal extraction stages reveals that pretreatment contributes 18–26% of total human health impacts for NMC811 cells (Fig. 3b). For LFP cells, this contribution increases to 23–36% (Supplementary Table 13), primarily because the subsequent metal extraction stage exhibits ~40% lower footprint. This reduction stems from the absence of nickel and cobalt in LFP cathodes, eliminating energy- and reagent-intensive steps such as solvent extraction, evaporation and crystallization. Regional context is another key factor influencing the footprint. Among major LIB markets—China, Europe and the USA[45]—pretreatment conducted in China shows 14–32% higher health impacts (Supplementary Table 13), largely due to its carbon-intensive energy mix. Considering variations in pretreatment routes, cathode chemistries and regions, the pretreatment stage contributes 16–38% of total human health impacts in hydrometallurgical recycling.

Among the pretreatment routes, chemical pretreatment exhibits the highest human health impact (Fig. 3c), mainly due to its complex operations, including solvent soaking that expends NMP, distillation that consumes energy and generates VOCs and additional air pollutant control required to remove nitrogen oxides (NO_x) originating from NMP (Supplementary Table 6). The impact is followed by thermal and mechanical routes, reaching 1,258, 887 and 746 $\times 10^{-6}$ disability-adjusted life years (DALY) per ton of processed NMC811 cells (abbreviated as t-cell hereafter), respectively. The primary components of health damage across all routes are global warming (42–59%) and particulate matter (PM) formation (27–44%). Figure 3d illustrates

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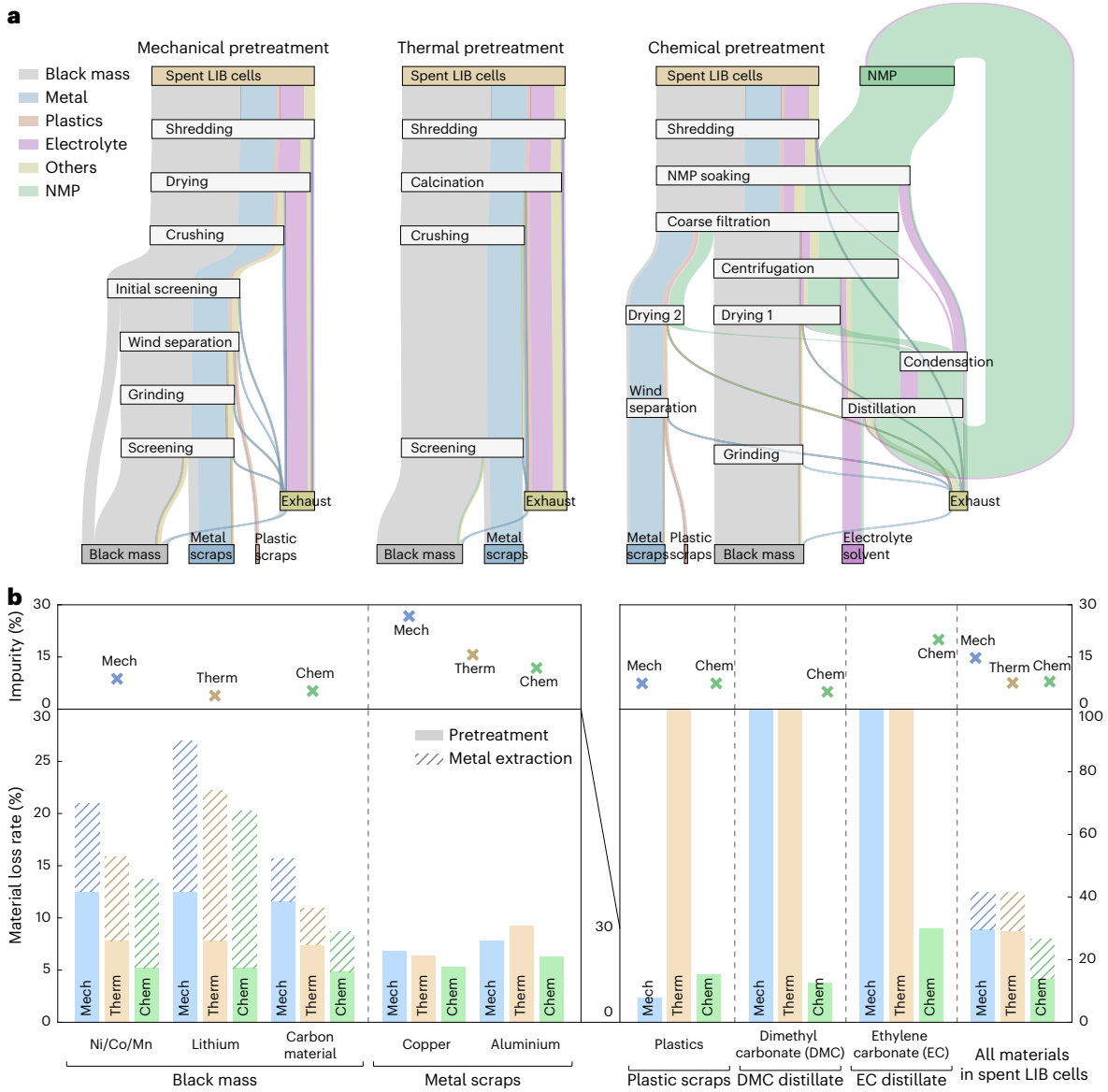


Fig. 2 | Material flows and losses across spent LIB pretreatment routes. **a**, Quantitative material flows of main materials (black mass, metal, plastics and electrolyte solvent) from spent NMC811 cells through individual process units in mechanical (Mech), thermal (Therm) and chemical (Chem) pretreatment routes (see Supplementary Tables 4–6 for other cathode chemistries). Partial recovery of black mass occurs via the dust collector hopper discharge. **b**, Material loss rates and product impurities from spent NMC811 cells across pretreatment routes, compared with the downstream metal extraction stage in hydrometallurgical recycling. Material losses include exhausts generated during pretreatment operations as well as materials presenting as impurities in other product streams. Corresponding BoMs across different routes and cathode chemistries are provided in Supplementary Tables 7–9. The primary recovery target is black mass, with metal scraps, plastic scraps and distillate as secondary products.

that, despite variations in cathode chemistries, pretreatment routes consistently follow the same increasing footprint order: mechanical, thermal, chemical. Mechanical pretreatment exhibits the greatest variation in health impacts across chemistries, with an ~42% difference (526–746 × 10^{-6} DALY per t-cell), largely driven by differences in carcinogenicity. These impacts stem from airborne PM generated during several size-reduction and separation steps, with nickel compounds serving as the primary driver[46], while nickel-free chemistries such as LFP, LMO and LCO show negligible carcinogenic impacts. Notably, although chemical pretreatment produces the least PM in exhaust streams (Supplementary Table 9), it incurs the highest PM formation-related health burden (~550 × 10^{-6} DALY per t-cell), more than double that of other routes. This is mainly attributed to the NMP solvent usage, whose upstream production carries substantial environmental impacts.

Figure 4a illustrates the environmental impacts of hydrometallurgical recycling and its pretreatment stage across 18 midpoint cause–effect categories. The analysis reveals that chemical pretreatment generally exhibits higher impacts than other routes, particularly for freshwater and marine eutrophication (FEU and MEU) and ozone formation affecting human health and terrestrial ecosystems (OFH and OFT), each contributing over 40% of total recycling impacts. These burdens are largely driven by NMP use, whose upstream production releases NO_x and VOCs to air and chemical oxygen demand (COD), phosphate and ammonium to water. By contrast, mechanical and thermal routes dominate terrestrial and marine ecotoxicity (MEC and TEC), contributing over half of entire recycling impacts. This is mainly due to the airborne emissions of copper, nickel and cobalt compounds, released alongside dust during LIB pretreatment. Among these metals, nickel compounds strongly influence human carcinogenic toxicity (HCT)[46].

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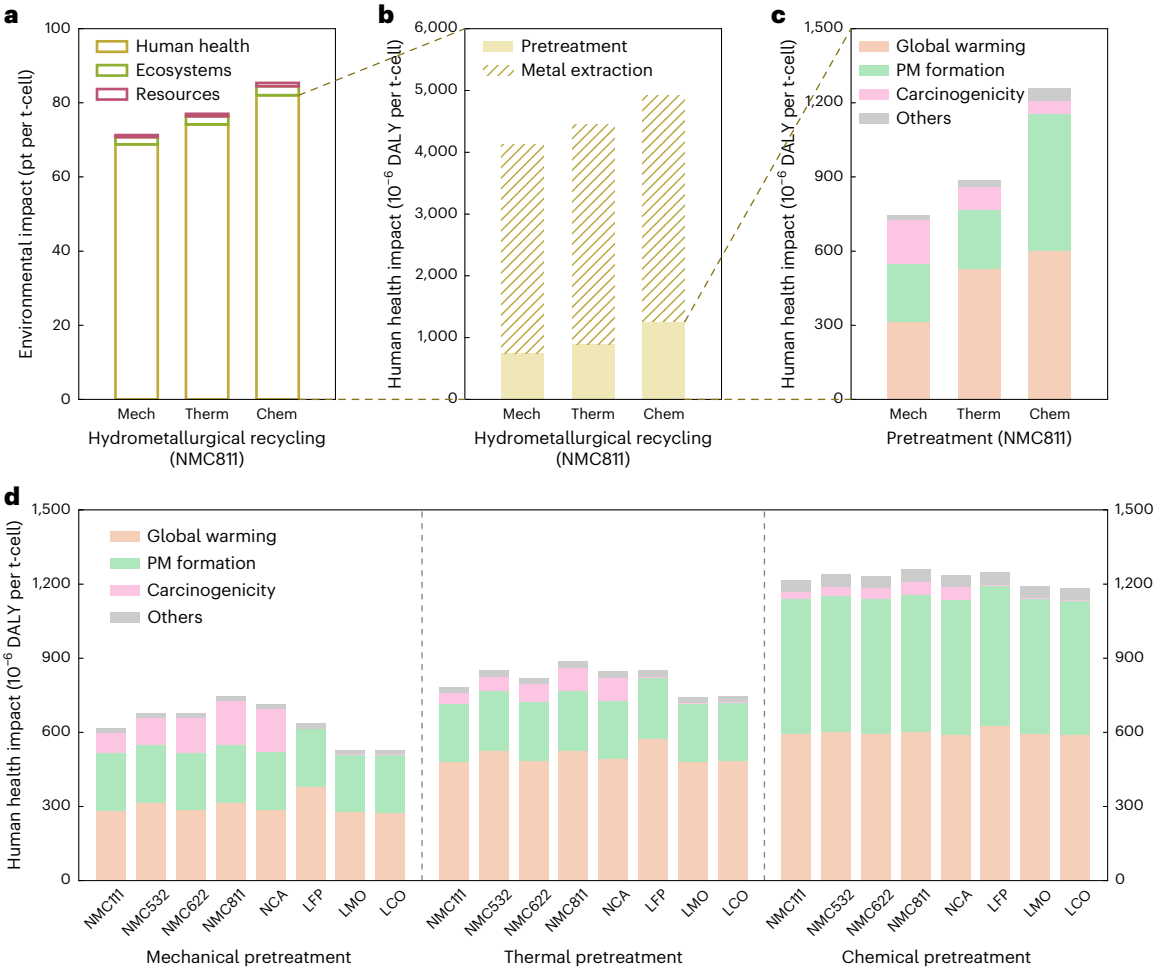


Fig. 3 | Endpoint environmental impacts of hydrometallurgical recycling and its pretreatment stage. Endpoint impacts represent final consequences of environmental damage on three areas of protection—human health, ecosystems and resources—assessed by the ReCiPe 2016 EndPoint (H/A) method on a global basis (see Supplementary Tables 13 and 14 for full-spectrum endpoint impacts across China, Europe and the USA). **a**, Single-score endpoint impacts per ton of NMC811 cells for hydrometallurgical recycling pathways using different pretreatment routes, with three damage categories aggregated and weighted in points (pt). **b**, Human health impacts per ton of NMC811 cells from the pretreatment stage and the subsequent metal extraction stage, measured in DALY that represent the loss of healthy lifespan. **c**, Contributions of principal categories to human health damage per ton of NMC811 cells across pretreatment routes. Results associated with **a**–**c** for LFP cells are provided in Supplementary Fig. 1. **d**, Contributions of principal categories to human health damage per ton of LIB cells across pretreatment routes and cathode chemistries.

For high-nickel chemistries (for example, NMC811 and NCA), mechanical pretreatment contributes three-quarters of total HCT, whereas for LFP cells this contribution falls below 10% (Supplementary Fig. 2).

Figure 4b,c quantify the carbon footprint and cumulative energy demand (CED) of the pretreatment stage. These indicators follow the same increasing order as human health impacts: mechanical, thermal and chemical. Carbon footprints ranges are 295–408, 513–617 and 625–666 $kgCO_2e$ per t-cell, with corresponding energy demands of 1,814–2,035, 3,699–4,076 and 9,195–9,682 MJ per t-cell, respectively. Among cathode types, LFP pretreatment shows the highest values for both indicators, mainly because of its higher electrolyte content (~19% of cell mass versus 12–15% for other chemistries; Supplementary Table 2), which increases energy use and CO_2 emissions during electrolyte decomposition. Regional differences are also evident, with pretreatment in China exhibiting 11–19% higher carbon footprints than in Europe and the USA (Supplementary Table 17). At the full hydrometallurgical recycling level, processing NMC811 cells results in 1,144–1,857 $kgCO_2e$ per t-cell across regions (that is, 3.3–7.4 times higher than waste-to-energy incineration per ton of municipal solid waste[47]). Pretreatment alone contributes 25–40% of this footprint, increasing to 35–56% for LFP cells as a result of their lower downstream metal extraction burdens (Supplementary Fig. 2). Similarly, pretreatment accounts for 11–49% of total energy demand, depending on pretreatment route, cathode chemistry and region. These findings highlight the critical role of pretreatment in determining the environmental performance of LIB recycling, indicating the importance of mitigating its footprint.

Process upgrades potentially halve pretreatment footprint

Environmental footprints (presented by human health impacts) across unit operations and substances in the pretreatment stage are illustrated in Fig. 5a for NMC811 cells (see Supplementary Tables 20 and 21 for other cathode chemistries). In mechanical and thermal routes, heating-intensive operations dominate footprints: drying in mechanical pretreatment contributes 43% of total baseline impacts, while calcination in thermal pretreatment accounts for 74%. The impacts largely stem from the VOC generation, which burdens downstream exhaust treatment systems. Implementing VOC recovery systems (for example, multistage condensers) in these routes could reduce VOC-related impacts by 82% and 58%, respectively. PM generation is another major contributor, representing 44% (mechanical) and 20% (thermal) of total baseline impacts, mainly originating

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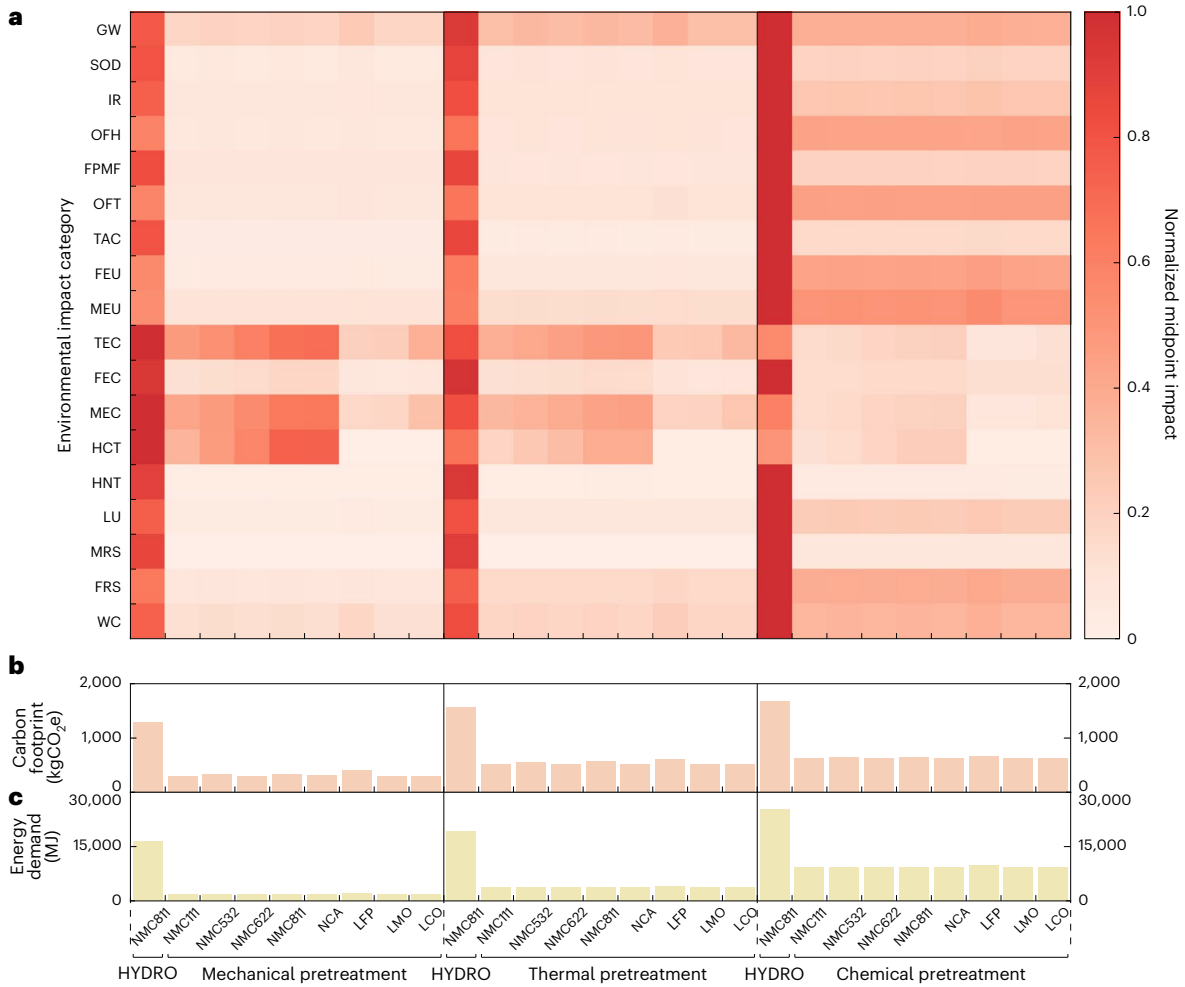


Fig. 4 | Midpoint environmental impacts, carbon footprint and energy demand of LIB hydrometallurgical recycling and its pretreatment stage. Midpoint impacts represent 18 cause–effect categories leading to endpoint damage, assessed by the ReCiPe 2016 MidPoint (H) method on a global basis. Full definitions of the midpoint impact category abbreviations are provided in Methods. Carbon footprint and CED are assessed by the IPCC 2021 GWP100 and CED methods, respectively. Full-spectrum impacts across regions are provided in Supplementary Tables 15–18. All panels share the same *x* axis. **a**, Normalized midpoint impacts comparing full hydrometallurgical recycling (applied to NMC811) with its pretreatment stage (applied across eight cathode chemistries) using mechanical, thermal or chemical routes. Values are normalized within each category using the maximum normalization method and visualized by the colour scale. **b**, Carbon footprint per ton of spent LIB cells processed through hydrometallurgical (HYDRO) recycling (NMC811) and its pretreatment stage (eight cathode chemistries). **c**, CED per ton of spent LIB cells processed through hydrometallurgical recycling (NMC811) and its pretreatment stage (eight cathode chemistries). Results for full hydrometallurgical recycling of LFP cells are provided in Supplementary Fig. 2.

from carcinogenic nickel-containing particulates released during physical processing (for example, shredding and crushing). Enhancing equipment sealing and workshop containment could mitigate PM-related impacts by over one-third. Thermal pretreatment also exhibits greater reliance on natural gas (NG), constituting 13% of total baseline impacts. Increasing heating device efficiency in thermal pretreatment through improved insulation and waste heat recovery could reduce NG-related impacts by 34%. In chemical pretreatment, the NMP soaking process dominates total baseline impacts (accounting for 57%), largely driven by solvent consumption. Enhancing solvent recovery using an additional vacuum distillation column could mitigate solvent-related impacts by 20%, while substituting NMP with alternative soaking media, such as DES[25,27], could further reduce remaining solvent impacts by 70%.

Several prominent cathode materials with distinct chemical compositions—NMC811, LFP and LCO—are selected to assess potential reductions in human health impacts (Fig. 5b). By implementing proposed process upgrade strategies, baseline pretreatment impacts could be mitigated by 35–50% across routes and cathode chemistries, resulting in residual impacts of 300–735 × 10^{-6} DALY per t-cell. Corresponding reductions in carbon footprint and energy demand reach 38–66% and 13–43%, respectively (Supplementary Figs. 3 and 4). These improvements translate to an 8–15% mitigation in total human health impacts across the full hydrometallurgical recycling pathway, alongside decreases of 15–26% in carbon footprint and 2–21% in energy demand. Among pretreatment routes, mechanical pretreatment offers the greatest reduction potential (Fig. 5b), achieving 43–50% decreases across cathode types, compared with 35–37% for thermal and 40–43% for chemical routes. Among various upgrade strategies, VOC recovery is the most effective for mechanical and thermal pretreatment, reducing total impacts by 29–43% and 24–29%, whereas alternative solvent use proves most impactful for chemical pretreatment, achieving 26–29% decreases. By contrast, improving energy efficiency in heating devices yields limited gains for mechanical and chemical pretreatment. The benefits of improved equipment sealing are route- and cathode-dependent; for example, it reduces impacts by 15% for mechanical pretreatment of high-nickel NMC811 cells but offers minimal gains for chemical pretreatment of LFP and LCO.

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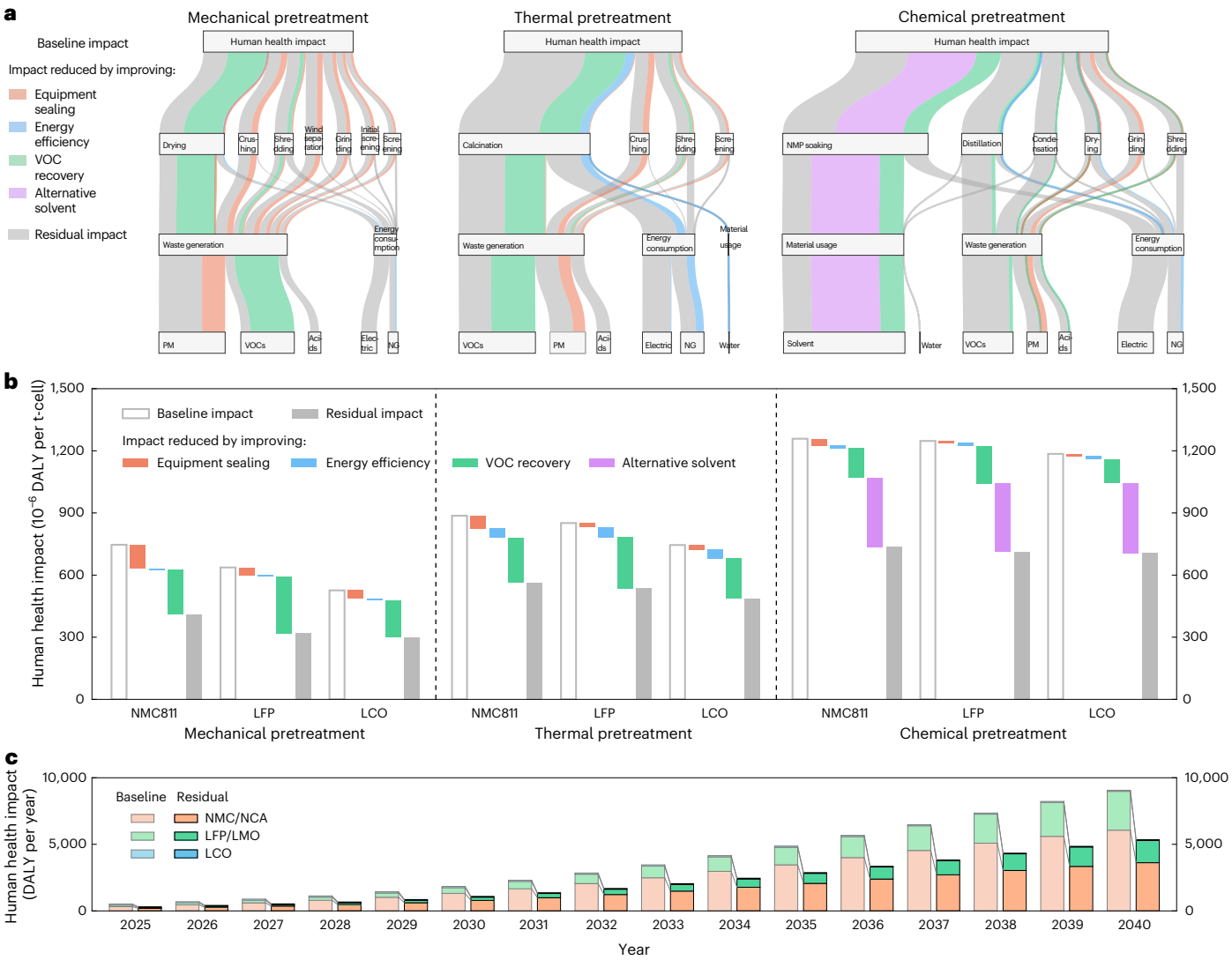


Fig. 5 | Environmental impact contributors and mitigation potentials through process upgrades across different pretreatment routes. Environmental impacts of contributors include burdens imposed on downstream exhaust treatment systems. Process upgrade strategies are assessed across four dimensions: equipment sealing, energy efficiency, VOC recovery and alternative solvent use. **a**, Human health impact flows and associated mitigation potentials from process upgrades across pretreatment routes for NMC811 cells, mapped at the process unit, operational behaviour and substance levels (see Supplementary Tables 20 and 21 for other cathode chemistries). **b**, Quantitative mitigation potentials of human health impacts from process upgrades across pretreatment routes for NMC811, LFP and LCO cells. **c**, Projected annual human health impacts from global spent LIB pretreatment operations during 2025–2040 under baseline and improved scenarios. Associated results for carbon footprint and CED are provided in Supplementary Figs. 3 and 4, respectively. NMC/NCA cathode category consists of NMC111, NMC532, NMC622, NMC811 and NCA.

Figure 5c presents projected annual human health impacts of global spent LIB pretreatment operations, along with the mitigation potential through process upgrades. Following the trend of EoL LIB volumes during 2025–2040 compiled in this study (Supplementary Table 25), baseline operational impacts are expected to increase 17-fold from 519 DALY yr^{-1} to 9,060 DALY yr^{-1}, while implementing process upgrades could mitigate these impacts by ~40%. Over the 15-yr period, cumulative impacts could decrease from 60,900 DALY to 36,000 DALY, equivalent to preserving the full healthy lifespan of over 300 individuals (assuming an 80-yr life expectancy). Among cathode chemistries, NMC/NCA category dominates global impacts, contributing 61–68% because of their prevalence in EVs and high retirement volumes. By contrast, the contribution from LCO, primarily used in consumer electronics, declines from 18% to negligible levels over the period as EV adoption accelerates. In terms of carbon footprint and energy demand, baseline annual values are projected to rise 17-fold from 0.29 to 5.08 $MtCO_2e$ yr^{-1} and from 2.9 to 49.7 PJ yr^{-1}, respectively (Supplementary Figs. 3 and 4). Process upgrades could mitigate these by ~46% and ~33%. Cumulatively, this translates to reductions from 34 $MtCO_2e$ to 18 $MtCO_2e$ and from 334 PJ to 223 PJ over the 15-yr period.

Discussion

This study presents an operational unit-level anatomy of mechanical, thermal and chemical pretreatment routes, demonstrating that pretreatment should not be generalized as a minor or uniform contributor to materials recovery and environmental impacts in LIB recycling. With the European Union targeting a 70% recycling rate of LIB materials by 2030[48], achieving high recovery efficiency at the pretreatment stage becomes essential. Among the three evaluated routes, chemical pretreatment offers superior recovery of black mass and metal scraps through fluid-based operations, while also inherently enabling electrolyte recovery along with NMP solvent recovery. By contrast, mechanical and thermal routes require additional equipment for electrolyte recovery, raising cost–benefit concerns. Our analysis shows that chemical

pretreatment combined with subsequent metal extraction achieves ~74% overall materials recovery, meeting regulatory targets, whereas adopting mechanical or thermal pretreatment decreases that value to ~58%. Chemical pretreatment could also produce black mass with lower impurity levels, ensuring efficient downstream metal extraction. Apart from material circulation, environmental impact represents another important dimension of sustainability. Under the cutoff LCA boundary applied in this study, we reveal that chemical pretreatment exhibits the highest operational environmental impacts among the routes. However, when evaluated using a consequential LCA boundary, it could also offer greater avoided impacts from higher recovery yields by substituting their upstream raw material production. Such net impact assessments should account for differences in cathode chemistries (for example, NMC and LFP), which have substantially different upstream production burdens that determine the magnitude of avoided impacts through their recycling activities. This future research initiative, comparing environmental impacts between 'urban mining' and natural resources extraction, can be supported by our foundational unit-level BoM datasets provided herein. Given its higher materials recovery and the promising application of green solvents, chemical pretreatment is likely to achieve the lowest net environmental impact under the consequential LCA framework.

Although chemical pretreatment shows the most promising performance in both materials recovery and net environmental impact, its industrial adoption largely depends on economic feasibility. While this route generates high revenues from recovered materials, our process inventory and BoMs indicate that it also entails the greatest operational complexity. The need for extensive manufacturing and exhaust treatment systems increases both capital and operating costs, including labour and reagent consumption. However, economic viability could be enhanced in regions such as China, where labour costs are lower than in Europe and the USA. Recent declines in NMP prices and the development of low-cost green solvents[25,27], along with potential policy incentives tied to the sustainability advantages of the route, could further improve feasibility. Economic performance also depends strongly on cathode chemistry. For example, chemical pretreatment is well-suited for Ni/Co-rich LIBs such as NMC, where recovery yields have high market value. By contrast, for Ni/Co-deficient chemistries, particularly LFP, chemical pretreatment and subsequent metal extraction may face economic challenges[49]. Given the large share of the LIB market for LFPs in China, China could encourage echelon use in less demanding applications (for example, stationary storage or low-speed EVs) as a favourable alternative, leveraging the long lifespan and safety advantages of LFPs[50]. For unqualified LFP cells that must enter traditional recycling streams, mechanical pretreatment is recommended because of its lower capital requirements and operational environmental impacts. This approach also aligns with the labour context of China, as solid-based processing tends to be less automated. Future economic assessments across pretreatment routes should account for cathode chemistry, regional–temporal market dynamics and technological advancements to warrant their industrial applications. Our foundational process inventory and BoM datasets can support such research initiatives to select appropriate pretreatment routes that balance the overall material–environmental–economic trade-offs.

Apart from guiding pretreatment route selection, this study also provides actionable insights for stakeholders designing or retrofitting processes in alignment with local regulatory, environmental and economic contexts. Across all pretreatment routes, the importance of VOC recovery should be highlighted. In mechanical and thermal pretreatment, integrating VOC recovery systems could effectively mitigate operational environmental impacts while enabling recovered electrolyte mixtures to be repurposed as liquid fuels[51], creating potential revenue streams to offset added capital investments. For chemical pretreatment, enhancing existing VOC recovery systems can further purify electrolytes separated from NMP solvent and reduce reagent consumption. Route- and cathode-specific process upgrades are also essential. For mechanical pretreatment of nickel-rich LIBs (for example, NMC811 and NCA), improving equipment sealing should be prioritized. These relatively low-cost measures are critical for further controlling fugitive emissions of carcinogenic particulates generated during physical processing. Unlike globally distributed impacts, such as climate change, such localized emissions could disproportionately increase occupational health risks for onsite workers. This necessitates high-standard equipment sealing and government-mandated PM concentration monitoring. Importantly, these impacts cannot be captured by some single-issue indicators such as carbon footprint or CED, underscoring the need for comprehensive, multi-impact assessments to guide regulatory decisions on industrial practices. For energy-intensive thermal pretreatment, energy efficiency upgrades, such as advanced heating technologies, improved insulation and waste heat recovery, can effectively deliver economic benefits, particularly in regions with high NG costs such as Europe. Although these upgrade measures require upfront investment, they yield long-term environmental–economic advantages. Again, our foundational datasets can support further techno–economic assessments to evaluate the cost–benefit trade-offs of such improvements and guide investment decisions. Where marginal returns discourage adoption, targeted government incentives could help align industrial behaviour with broader sustainability goals. Finally, alongside active process improvements, passive gains from cleaner national energy mixes have also been identified as important. Achieving sustainable EoL LIB management will require coordinated efforts across technological innovation, economic policy and environmental regulation.

Our unit-level simulations are designed to represent industrial-scale LIB pretreatment, incorporating factory configurations such as parallel production line setups, equipment models matched to line capacity and tailored exhaust treatment systems compliant with air emissions regulations (Methods). These features ensure applicability across industrial scales by accommodating different numbers of production lines. The resulting BoM datasets and interpreted outcomes are validated against enterprise disclosures and existing literature, demonstrating overall consistency while clearly explaining discrepancies (Methods). The inherent strength of our high-resolution, bottom-up simulations, combined with external validation, secures the reliability of our modelling framework, BoM datasets and analytical results, as well as their practical applications spanning academic research, industrial practice and policy decision-making. However, we also acknowledge certain limitations. Despite extensive data collection, obtaining complete and up-to-date information for all process units remains challenging, considering corporate confidentiality. We transparently document assumptions, conduct prospective scenario analyses on key parameters and disclose data validation procedures, all of which help to mitigate potential inaccuracies in our simulations. On the other hand, we model only one representative configuration for each pretreatment route using a typical set of LIB composition data. The diversity of industrial process variants and evolving LIB chemistries could therefore introduce inconsistencies, uncertainties or delays in data accuracy. Similarly, regional differences in air emissions regulations may require varying levels of exhaust treatment system intensity. Recognizing these limitations not only contextualizes our findings into practical applications but also identifies promising directions for refining sustainable EoL LIB management strategies to reconcile academia–government–industry collaborations.

Methods

Operational unit-level modelling at industrial scale

The performance simulation of the pretreatment system is built around a model of an industrial-scale facility designed to process spent LIB cells into black mass. This facility includes a single dedicated workshop configured for one of three pretreatment routes—mechanical, thermal

or chemical—applied to one of eight LIB cathode chemistries—NMC111, NMC532, NMC622, NMC811, NCA, LFP, LMO and LCO. The workshop in the facility is configured with two parallel production lines, each capable of processing 1.5 t of spent LIB cells per hour, in line with industrial practices. This workshop incorporates an exhaust gas treatment system to ensure compliance with environmental regulations, capturing emissions from both the workshop environment and the process equipment. The facility is modelled to operate continuously for 24 h per day over 300 days per year, resulting in an annual processing capacity of 30 kt of LIB cells, corresponding to a capacity factor of 0.72. All three pretreatment routes (mechanical, thermal and chemical) start with initially shredding spent LIB cells into mixed fragments for size reduction, enabling the operation of their respective key process units (grinding, calcination and solvent soaking). Detailed operational mechanisms of manufacturing units and exhaust treatment units can be found in Supplementary Note 2. To ensure a fair comparison, when a single process unit is used across multiple pretreatment routes, such as initial shredding for size reduction in all three routes, the working conditions and parameter settings are kept consistent. Among all the parameter settings, several are important that could influence the environmental impacts of the overall pretreatment routes. These include the exhaust collection efficiency from both processing devices and workshops, the thermal efficiency of heating units, the VOC recovery rate in the corresponding recovery systems and alternative solvent usage. Optimistic assumptions of these parameters are integrated in process upgrade strategies for prospective scenario analysis, serving as process upgrade strategies (Supplementary Note 8).

Operational parameters across process units, such as device-specific energy consumption, are primarily derived from commercial sources, including trade platforms such as Alibaba and Made-in-China, where equipment manufacturers provide specifications such as rated power and processing capacity for various models. On the basis of the factory configuration assumed in this study, appropriate equipment models are selected to evaluate the energy consumption for each process unit. Theoretical parameters, such as stoichiometric conversion factors, are calculated from chemical reaction equations. For instance, the conversion of VOCs to CO_2 is determined on the basis of the carbon content of the source compounds, while the decomposition of $LiPF_6$ to LiF is based on its chemical reaction pathway. Empirical parameters, including the PM generation rate from black mass, delamination efficiency of black mass from metal foils and air pollutant removal efficiency, are primarily obtained from environmental impact assessment (EIA) reports and peer-reviewed literature. In cases where empirical data are unavailable, estimates and assumptions are made on the basis of relative relationships informed by existing studies and our practical experience. For example, the PM generation rate from black mass is estimated to be ten times higher than that from metal scraps during mechanical abrasion, based on factory-monitored data[52]. Similarly, the separation efficiency of Cu, Al and plastics from black mass during screening is assumed to follow the descending order of Cu > Al > plastics, reflecting their relative brittleness and tendency to form fine particles upon impact. Additionally, fluid-based separation processes (for example, coarse filtration) are assumed to achieve higher separation efficiencies than solid-based methods (for example, screening), because of reduced material agglomeration and lower risk of equipment clogging. Further details on assumptions and estimation methods for missing parameters are provided in Supplementary Table 1.

Mass and energy balance simulations

On the basis of the configured pretreatment system, mass and energy balance simulations are conducted at the level of each individual process unit. For each process unit, the following parameters are calculated: (1) inflow and outflow of material mixtures, (2) energy and auxiliary material consumption and (3) generation of exhaust streams. The simulations resolve the material flows and exhaust flows down to specific species—including metal foils (Al and Cu), electrolyte solvents (EC and DMC) and plastic polymers such as polypropylene (PP), polyethylene (PE) and polyethylene terephthalate (PET). Energy demand is calculated for both thermal and electrical processes. Thermal energy use is estimated on the basis of the thermophysical properties of each species, including specific heat capacity, enthalpy of vaporization, fusion and decomposition, as provided in Supplementary Table 3. These parameters, combined with device-specific energy efficiency values, are used to estimate the NG input required for heating devices. Electrical energy use for each piece of equipment is primarily determined by its rated power and throughput capacity. For induced draft fans used in exhaust gas ventilation, energy consumption is modelled on the basis of required airflow rates that ensure compliance with regulatory limits on pollutant concentrations while maintaining effective system operation. The relevant estimation formulas of mass and energy balance are provided in Supplementary Notes 3 and 4. Following the step-by-step mass and energy balance simulations in the pretreatment stage, the detailed material and energy flow data, as well as summarized BoM datasets, are compiled in Supplementary Tables 4–9.

While the BoM datasets for the LIB pretreatment stage are derived from detailed operational unit-level simulations, these data for the subsequent metal extraction stage are compiled from a range of sources, including peer-reviewed studies and enterprise disclosures. These references include data obtained from industrial investigations as well as simulation-based assessments, with the majority published within the past 5 years. Owing to limitations in data availability, the compilation focuses on two cathode chemistries: NMC811 and LFP. These data and associated references are detailed in Supplementary Table 10. By combining the BoM data from both the pretreatment and metal extraction stages, complete hydrometallurgical recycling pathways are constructed for NMC811 and LFP cells using various pretreatment routes. These integrated BoM datasets are presented in Supplementary Table 11.

Materials recovery and environmental impact assessment

Using the BoM datasets, we conduct a systematic evaluation of materials recovery performance for each pretreatment route across all LIB chemistries. The recovery rate of each material is calculated as the mass retained in products relative to its original mass in the spent LIB cells. Materials that are not recovered are classified as losses, which can either be emitted in exhaust streams generated during pretreatment or retained as impurities within other products. To account for the influence of impurities on downstream processes, we introduce an impurity penalty in the pretreatment stage. This penalty reflects a reduction in critical metal recovery during the subsequent metal extraction stage, based on the presence of impurities in the black mass. Related equations are detailed in Supplementary Note 6. The overall materials recovery rate of LIB cells during pretreatment is defined as the mass of target materials in the products divided by the total initial mass of LIB cells. In the metal extraction stage, recovery is determined by the mass of metal ions originating from the black mass, excluding external reagent contributions such as sulfate ions in $CoSO_4$, divided by the mass of the pure fraction of black mass recovered during pretreatment. The oxygen in cathode materials such as $Li(Ni_{0.33}Mn_{0.33}Co_{0.33})O_2$ is considered totally lost as H_2O during the leaching process.

Using the material and energy flow data for each individual process unit, along with the summarized BoM datasets, a comprehensive LCA is carried out for the pretreatment stage. Operational environmental impacts are evaluated throughout this study by using a cutoff LCA boundary, accounting for system inputs (for example, energy, raw materials and water) and output waste streams (for example, emissions, industrial waste and wastewater), while excluding avoided impacts from recovered black mass and byproducts as well as upstream burdens embodied in spent LIB cells. The functional unit for the assessment

is defined as the pretreatment of 1 t of spent LIB cells into black mass within the modelled factory system, applying mechanical, thermal or chemical pretreatment routes. LCA modelling is conducted using SimaPro v.9.5.0.2, referencing the Ecoinvent 3 database (allocation at point of substitution—unit [APOS, U]). Infrastructure-related processes and long-term emissions are excluded from the analysis. The inventory data are set to different location bases, covering global, China, Europe and the USA, to evaluate the influence of regional variations on environmental impacts. To estimate global-scale environmental impacts from spent LIB pretreatment activities, the annual volume of retired LIB cells by region and cathode chemistry is projected, detailed in Supplementary Note 9.

Midpoint impacts refer to specific environmental problems or processes that occur along the environmental cause–effect chain[46]. They represent measurable points where pollutants or resource uses exert pressure on the environment, such as global warming, acidification, eutrophication or ozone depletion, assessed by the ReCiPe 2016 MidPoint (H) method[46]. The assessed midpoint impact categories include global warming (GW), stratospheric ozone depletion (SOD), ionizing radiation (IR), ozone formation (human health) (OFH), fine particulate matter formation (FPMF), ozone formation (terrestrial ecosystems) (OFT), terrestrial acidification (TAC), freshwater eutrophication (FEU), marine eutrophication (MEU), terrestrial ecotoxicity (TEC), freshwater ecotoxicity (FEC), marine ecotoxicity (MEC), human carcinogenic toxicity (HCT), human non-carcinogenic toxicity (HNT), land use (LU), mineral resource scarcity (MRS), fossil resource scarcity (FRS) and water consumption (WC). By contrast, endpoint impacts reflect the final consequences of these environmental mechanisms on areas of protection, that is, human health, ecosystems and natural resources[46]. These endpoint categories capture the actual damage or risks posed to these critical protection areas, based on the cumulative effects along the entire cause–effect chain. The three damage categories are quantified, normalized, weighted and aggregated into a single score of overall endpoint environmental impact, using ReCiPe 2016 EndPoint (H/A) method[46]. Similarly, a total of 18 midpoint impact categories, each contributing to one or more of the three endpoint damage categories, are evaluated through the same process. For example, eight midpoint impact categories, such as global warming and PM, are quantified, normalized, weighted and then aggregated into the overall impact on human health damage, measured in DALY. According to a review article[53], this is the recommended approach for performing and reporting the LCA using the ReCiPe method. For single-issue environmental indicators assessed in this study, that is, the carbon footprint and energy demand, the IPCC 2021 GWP100 and CED methods are used, respectively.

Validation of datasets and analytical results

To verify the robustness and industrial relevance of the operational unit-level modelling developed for the pretreatment stage, we conduct a series of validation comparisons using data disclosed by commercial enterprises and published in peer-reviewed literature. These comparisons demonstrate that the simulated BoM datasets, as well as their associated results, show consistency with real-world operations, supporting their reliability for industrial-level applications. A key focus of the validation involves energy consumption, one of the most critical inputs in the BoM for the pretreatment stage. For mechanical pretreatment, AkkuSer in Finland reports energy use of 300 kWh per t-cell using their technology[54], while Machala et al.[10], who investigate Redwood Materials in the USA, report a value of 290 kWh per t-cell (after normalizing the literature data to the same functional unit of processing 1 t of LIB cells; Supplementary Note 5). Our modelled energy consumption falls within the range of 225–245 kWh per t-cell across LIB chemistries (Supplementary Table 7). In the case of thermal pretreatment, Machala et al.[10] report a consumption of 514 kWh per t-cell, whereas our modelled range is 566–620 kWh per t-cell (Supplementary Table 8). All these differences are within acceptable margins when considering variations in pretreatment route designs, such as the number of process units per function (for example, size reduction or separation), equipment variants and heating sources (for example, electricity or NG), as well as differences in evaluated system boundaries, including workshop overheads such as lighting, heating, ventilation and air conditioning. For chemical pretreatment, which remains largely at the laboratory research stage with limited industrial application, there is currently a lack of publicly available reference data. In this case, our high-resolution simulations can serve as a forward-looking benchmark for guiding future industrial implementation.

In addition to energy data, PM emissions represent another critical component in assessing the environmental performance of the pretreatment stage. An EIA report from a Chinese facility using mechanical pretreatment indicates that the PM generation rate of black mass can reach up to 8% (ref. 55). Our modelled generation rate of 6.3% (Supplementary Table 4) falls within an acceptable range, with deviations probably arising from specific differences in process configurations, such as the number and intensity of grinding and screening steps. Moreover, Jiang et al.[38] report collection of data from industry expert estimations and enterprise sources in China, estimating PM emissions into the air from mechanical pretreatment to be in the range of 735–769 gPM per t-cell, while our simulation results are in the range of 475–530 gPM per t-cell (Supplementary Table 7), again showing close alignment. These comparisons support the credibility of our emission modelling and the process flow assumptions underlying our BoM datasets. In terms of other airborne emissions, our modelled source-point emissions via chimney, derived from simulated material and energy flows, consistently fall under 70% of general regulation thresholds, which are set at 20, 60, 200, 5 and 30 mg m^{-3} for PM, VOCs, NO_x, HF and NH_3, respectively (Supplementary Tables 4–6). Collectively, these validations of fundamental BoM datasets affirm the applicability of our model for both academic research and industrial planning.

In addition to the BoM validation, we also verify the associated analytical results by comparing them with several external sources. From the materials recovery perspective, the LithoRec II project[56] and a peer-reviewed study[57] both report ~85% black mass recovery via industrial-scale physical processing, while a laboratory-scale investigation documents a cathode peel-off efficiency of 87% (ref. 58), consistent with our modelled 87–88% cathode recovery in the mechanical pretreatment route (Supplementary Table 12). Similarly, for thermal pretreatment, LithoRec II reports that over 91% black mass recovery could be achieved at the industrial scale[56] and a datasheet from the German company Accurec indicates a cathode recovery of 96% using their technology[59]. Our model estimates of 92–93% cathode recovery in the thermal pretreatment route fall well within this reported range. From the environmental impact perspective, Jiang et al.[38], drawing on enterprise data and expert estimations, report that the carbon footprint associated with mechanical pretreatment for NMC and LFP chemistries is in the range of 400–500 kg CO_2eq per t-cell (ref. 38). By comparison, our results are in the range of 295–408 kgCO_2e per t-cell across various cathode chemistries (Supplementary Table 18). Regarding the contribution of the pretreatment stage to the environmental impact of full hydrometallurgical recycling, Jiang et al.[38] estimate that mechanical pretreatment accounts for 15–43% of the total carbon footprint, whereas our analysis indicates a contribution of 25–46% (Supplementary Table 17). Although some variation exists between our results and those reported data, these differences remain within acceptable margins considering variations in process design.

However, we also recognize that notable inconsistencies exist between our findings and those reported in previous studies. For instance, Machala et al.[10], in the assessment of Redwood Materials, report that $PM_{2.5}$ formation during mechanical and thermal pretreatment of NCA cells ranged from 0 to 13 g$PM_{2.5}$e per t-cell and 3–23 g$PM_{2.5}$e

per t-cell, respectively, across various electricity grids in the USA. By contrast, our modelled results for NMC811 indicate substantially higher values, ranging from 299–376 $gPM_{2.5}e$ per t-cell to 269–386 $gPM_{2.5}e$ per t-cell across global, China, Europe and US regions (Supplementary Table 15). These order-of-magnitude differences are primarily attributable to disparities in the fundamental datasets used to quantify the environmental impacts of electricity generation. Specifically, Machala et al.[10] use electricity burden data from the GREET model[60], which estimates $PM_{2.5}$ formation from electricity generation at 0–40 $gPM_{2.5}e$ per MWh, depending on the grid mix in the USA. By contrast, our study draws from the Ecoinvent v.3.8 database, integrated within the SimaPro v.9.5.0.2 software, which provides values in the range 554–1,576 $gPM_{2.5}e$ per MWh across regional energy mixes. This emission factor difference could stem from the inclusion of more upstream burdens in the Ecoinvent database. Acknowledging and understanding these data source discrepancies, as well as ensuring their transparency, is vital for enabling cross-study comparisons and preventing misleading decision-making. In addition to differences in electricity inventory, variations in system boundaries also contribute to inconsistencies in the results. Our modelling scope extends beyond direct operational emissions to include the environmental burdens associated with exhaust gas treatment systems, such as the consumption of NaOH and activated carbon. Furthermore, we account for fugitive emissions occurring during exhaust collection at operation units and during the discharging, transfer and packaging of captured black mass from baghouse and cyclone hoppers (Supplementary Note 3).

Machala et al.[10] also report that the carbon footprint of mechanical and thermal pretreatment for NCA cells in the ranges of 11–198 $kgCO_2e$ per t-cell and 90–403 $kgCO_2e$ per t-cell, respectively. In comparison, our modelled results for NMC811 cells show higher values, in the ranges of 315–360 $kgCO_2e$ per t-cell and 529–596 $kgCO_2e$ per t-cell (Supplementary Table 17). Similarly, these differences primarily arise from variations in the fundamental data on electricity-related environmental burdens. Machala et al.[10] use electricity carbon intensity values of 37–682 $kgCO_2eq$ per MWh depending on the grid mix, while our study adopts values in the range of 375–1043 $kgCO_2e$ per MWh. Beyond the source data differences, pretreatment route designs could also contribute to the observed variations in carbon footprint between studies. These include the treatment method for electrolyte (for example, recovered or decomposed), the operational mechanism of the heating unit[10] and the energy source used for heating (for example, electricity or NG). Finally, Machala et al.[10] report that mechanical and thermal pretreatment for NCA cells accounts for 2–7% and 13–15%, respectively, to the carbon footprint of the full hydrometallurgical recycling. By contrast, our results for NMC811 cells indicate higher contributions of 25–28% and 35–38% for these two pretreatment routes. Apart from the previously discussed differences in reported carbon footprints from the pretreatment stage itself, the observed variations in the contributions of pretreatment are also driven by differences in the underlying BoM data used for the downstream metal extraction stage. As indicated in many studies, the carbon footprint of different hydrometallurgical recycling processes could vary by several folds[35,36,61]. While the values reported in ref. 10 are derived from processes implemented by Redwood Materials, our BoM dataset for the downstream metal extraction stage integrates a broad range of sources, including several company disclosures and peer-reviewed literature (Supplementary Table 10). Overall, the consistency between our estimates and available literature, along with a transparent explanation of any discrepancies, supports the robustness and credibility of our operational unit-level modelling and the associated simulation results for the LIB pretreatment stage.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The process inventories for different pretreatment routes, including manufacturing and exhaust treatment units, are summarized in Supplementary Table 1. Mass and energy flow data and summarized BoM datasets for various pretreatment routes and LIB chemistries are detailed in Supplementary Tables 4–9. The comparative BoM data on hydrometallurgical recycling pathways applied to NMC811 and LFP are in Supplementary Table 11. Full-spectrum endpoint- and midpoint-level impacts for hydrometallurgical recycling and associated pretreatment stages across LIB chemistries and regional contexts are presented in Supplementary Tables 13–16, with carbon footprint and energy demand results in Supplementary Tables 17 and 18. Impact breakdowns under baseline and improved scenarios appear in Supplementary Tables 20 and 21. Projected LIB production, retirement and cathode chemistry distributions across applications and regions are detailed in Supplementary Tables 23–25. Additional data supporting the modelling and findings of this study can be found in Supplementary Information.

Code availability

The LCA modelling is performed using SimaPro v.9.5.0.2 and the source files of this software, along with the Python code for raw data processing, are available in Supplementary Code.

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Author contributions

The initial design of this study originated from B.L. and V.W.-C.C., and V.W.-C.C. contributed further conception and expertise. B.L. performed this research and mainly conducted data collection, analysis and interpretation. J.Z. and L.Y. provided the resources required for modelling and simulation. L.Y., X.W., Z.S. and W.G. assisted with data collection and compilation. B.L. and V.W.-C.C. led the drafting of this paper. V.W.-C.C., B.L., J.Z., L.Y., X.W., Z.S. and W.G. contributed substantially to revisions.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41893-026-01767-1.

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Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☒	☐ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data are mainly collected from industry reports and academic literature. No code was used to collect the data
Data analysis	Life cycle assessments in this study were conducted using using SimaPro 9.5.0.2 (https://simapro.com). The setup process, including bills of materials and interpretation methods, is described in detail in the supplementary information. For further information, refer to the section titled "Note S7. Life Cycle Assessment". Additional algorithms used in this study, along with their corresponding equations, are also provided in the supplementary information file. The source files of this software, along with the Python code for raw data processing, are available in the Supplementary Code.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

This paper includes a Data Availability Statement. We declare that all necessary raw data and intermediate data required for further analysis are provided in the Supplementary Information.

The process inventories for different pretreatment routes, including manufacturing and exhaust treatment units, are summarized in Supplementary Table 1. Mass and energy flow data and summarized BoM datasets for various pretreatment routes and LIB chemistries are detailed in Supplementary Tables 4–9. The comparative BoM data on hydrometallurgical recycling pathways applied to NMC811 and LFP are in Supplementary Table 11. Full-spectrum endpoint- and midpoint-level impacts for hydrometallurgical recycling and associated pretreatment stages across LIB chemistries and regional contexts are presented in Supplementary Tables 13–16, with carbon footprint and energy demand results in Supplementary Tables 17 and 18. Impact breakdowns under baseline and improved scenarios appear in Supplementary Tables 20 and 21. Projected LIB production, retirement, and cathode chemistry distributions across applications and regions are detailed in Supplementary Tables 23–25. Additional data supporting the modeling and findings of this study can be found in the Supplementary Information.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N.A.
Reporting on race, ethnicity, or other socially relevant groupings	N.A.
Population characteristics	N.A.
Recruitment	N.A.
Ethics oversight	N.A.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This study develops high-resolution models of the LIB pretreatment stage at the level of individual operational units, designed to reflect industrial-scale practices. The pretreatment framework covers representative mechanical, thermal, and chemical routes, each paired with appropriate exhaust treatment systems. Every unit operation is parameterized, where mass and energy balance simulations are performed across each unit to allow detailed tracking of material flows and associated environmental burdens.
Research sample	The model is developed based on parameter settings for each unit operation. These include operational parameters (e.g., device-specific energy consumption), theoretical parameters (e.g., stoichiometric conversion factors), and empirical parameters (e.g., PM generation rate from black mass, delamination efficiency of black mass from metal foils, and air pollutant removal efficiency).
Sampling strategy	Operational parameters across process units are primarily derived from commercial sources, including trade platforms like Alibaba and Made-in-China, where equipment manufacturers provide specifications such as rated power and processing capacity for various models. Theoretical parameters are calculated from chemical reaction equations. Empirical parameters are primarily obtained from environmental impact assessment (EIA) reports and peer-reviewed literature. In cases where empirical data are unavailable, estimates and assumptions are made based on relative relationships informed by existing studies and our practical experience.
Data collection	Data are collected from mentioned sources manually. No code was used to collect the data.

Timing and spatial scale	Most of the data comes from literature published within the past five years. The parameter settings are based on global data, but when interpreting the environmental impact of a specific region, the relevant datasets (such as national energy structures) are adjusted according to local conditions.
Data exclusions	When collecting individual parameter data, if multiple sources report significantly differing values, priority is given to data published in prestigious journals within the past five years, while other data are discarded. When values are provided as acceptable ranges, the mean value is used.
Reproducibility	The analytical results can be reproduced by raw data and intermediate data provided in the Supplementary Information.
Randomization	N.A.
Blinding	N.A.

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	N.A.
Novel plant genotypes	N.A.
Authentication	N.A.